Applied Economics

Difference-in-Differences Estimator

Beate Thies

Winter 2024/25

Introduction

Motivation

Multiple regression often does not provide causal estimates because of **endogeneity**.

- observational data: subjects are not randomly assigned to treatment ($\neq$ experimental data)
- data reflects real life choices – often correlated with unobservable characteristics

Motivation

Multiple regression often does not provide causal estimates because of **endogeneity**.

- observational data: subjects are not randomly assigned to treatment ($\neq$ experimental data)
- data reflects real life choices – often correlated with unobservable characteristics

There are three widely used **quasi-experimental methods** that intend to *mimic the idea of random treatment assignment:*

- instrumental variables (IV)
- differences-in-differences (DiD)
- regression discontinuity designs (RDD)

Differences-in-Differences

The DiD approach exploits changes in a policy over time that don't affect everyone to estimate the causal effect of the policy.

Differences-in-Differences

The DiD approach exploits changes in a policy over time that don't affect everyone to estimate the causal effect of the policy.

You need data on 2 groups:

- **control group**: never receives treatment/policy
- **treatment group**: initially untreated and then receives treatment/policy

Differences-in-Differences

The DiD approach exploits changes in a policy over time that don't affect everyone to estimate the causal effect of the policy.

You need data on 2 groups:

- **control group**: never receives treatment/policy
- **treatment group**: initially untreated and then receives treatment/policy

You need data for 2 time periods: **before and after treatment**.
$\rightarrow$ Requires panel data.

Under certain assumptions, the control group can serve as the counterfactual for the treatment group. $\rightarrow$ **Difference in outcomes between groups after vs. before treatment** gives causal effect of interest.

Example: Causal effect of Minimum Wage on Employment

Card & Krueger (1994)

Why can we not just run a regression of employment rate on minimum wage (MW) at the state level?

Card & Krueger (1994)

Why can we not just run a regression of employment rate on minimum wage (MW) at the state level?

Card and Krueger (1994) use the DiD approach to estimate the effect of an increase in the MW on the employment rate in the fast-food industry.

- New Jersey (NJ) increased the state MW in 1992 from \$4.25 to \$5.05 per hour (treatment group)
- Neighboring Pennsylvania (PA) did not change its minimum wage level (control group)
- The authors surveyed 410 fast-food establishments in NJ and eastern PA, once before the NJ MW increase (early 1992), and once after the NJ MW increase (late 1992)

DiD: Idea

We observe how employment changed in NJ from before to after the increase in the MW.

Does this allow to infer about the causal effect of the MW increase? Why (not)?

DiD: Idea

We observe how employment changed in NJ from before to after the increase in the MW.
Does this allow to infer about the causal effect of the MW increase? Why (not)?

To identify the effect of the MW increase on employment, we need a counterfactual: How would employment have changed during that time period, *if MW had not been increased*?

DiD: Idea

We observe how employment changed in NJ from before to after the increase in the MW.
Does this allow to infer about the causal effect of the MW increase? Why (not)?

To identify the effect of the MW increase on employment, we need a counterfactual: How would employment have changed during that time period, *if MW had not been increased*?

Use time trend in neighboring PA (control group) as counterfactual: in the absence of the policy, we would have expected the same trend in NJ

- PA and NJ are very similar: similar institutions, similar habits, similar consumers, similar incomes, similar weather, etc.

DiD Estimator: Basic Form

Table 1: Sample Means of Employment (FTE) per Store

	NJ
Before Treatment	20.4
After Treatment	21.0
Difference (A-B)	0.6

DiD Estimator: Basic Form

Table 1: Sample Means of Employment (FTE) per Store

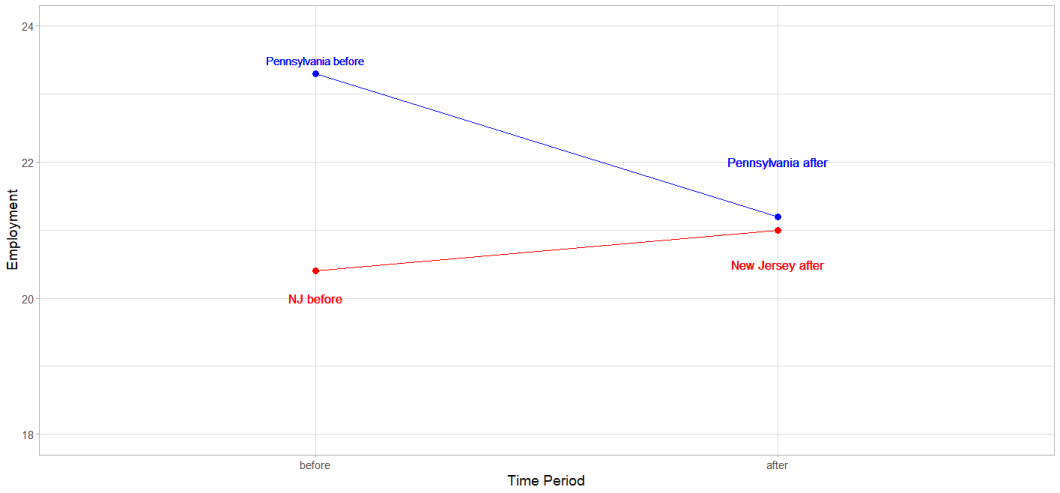
	NJ	PA
Before Treatment	20.4	23.3
After Treatment	21.0	21.2
Difference (A-B)	0.6	-2.1

DiD Estimator: Basic Form

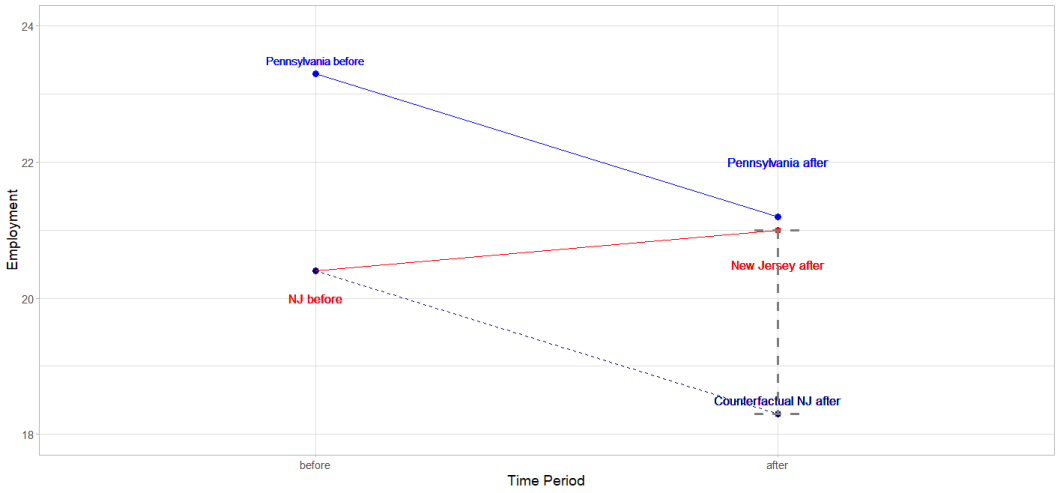
Table 1: Sample Means of Employment (FTE) per Store

	NJ	PA	
Before Treatment	20.4	23.3	
After Treatment	21.0	21.2	
Difference (A-B)	0.6	-2.1	DiD estimate: $0.6 - (-2.1) = 2.7$

DiD Estimator: Graphical



DiD Estimator: Graphical



Diff-in-Diff Estimator: Basic Form

DiD Estimator: Basic Form

More generally, the DiD estimator is given as:

$$\begin{aligned}\widehat{\beta}^{\text{DiD}} &= (\overline{Y}^{\text{treatment,after}} - \overline{Y}^{\text{treatment,before}}) - (\overline{Y}^{\text{control,after}} - \overline{Y}^{\text{control,before}}) \\ &= \Delta\overline{Y}^{\text{treatment}} - \Delta\overline{Y}^{\text{control}}\end{aligned}$$

with

$\overline{Y}^{\text{treatment,before}}$ sample average in the TG before the treatment $\rightarrow \overline{Employm}^{\text{NJ,before}}$

$\overline{Y}^{\text{treatment,after}}$ sample average in the TG after the treatment $\rightarrow \overline{Employm}^{\text{NJ,after}}$

$\overline{Y}^{\text{control,before}}$ sample average in the CG before the treatment $\rightarrow \overline{Employm}^{\text{PA,before}}$

$\overline{Y}^{\text{control,after}}$ sample average in the CG after the treatment $\rightarrow \overline{Employm}^{\text{PA,after}}$

Parallel Trends Assumption

The crucial assumption to identify the causal effect of interest with the DiD estimator is the **Common or Parallel Trends Assumption:**

In the absence of treatment, time trends in the treatment group would have been the same as in the control group.

Parallel Trends Assumption

The crucial assumption to identify the causal effect of interest with the DiD estimator is the **Common or Parallel Trends Assumption:**
In the absence of treatment, time trends in the treatment group would have been the same as in the control group.

Example: absent any minimum wage increase, Pennsylvania's fast-food employment trend would have been what we should have expected to see in New Jersey. $\rightarrow$ trend in PA is a reliable counterfactual for NJ

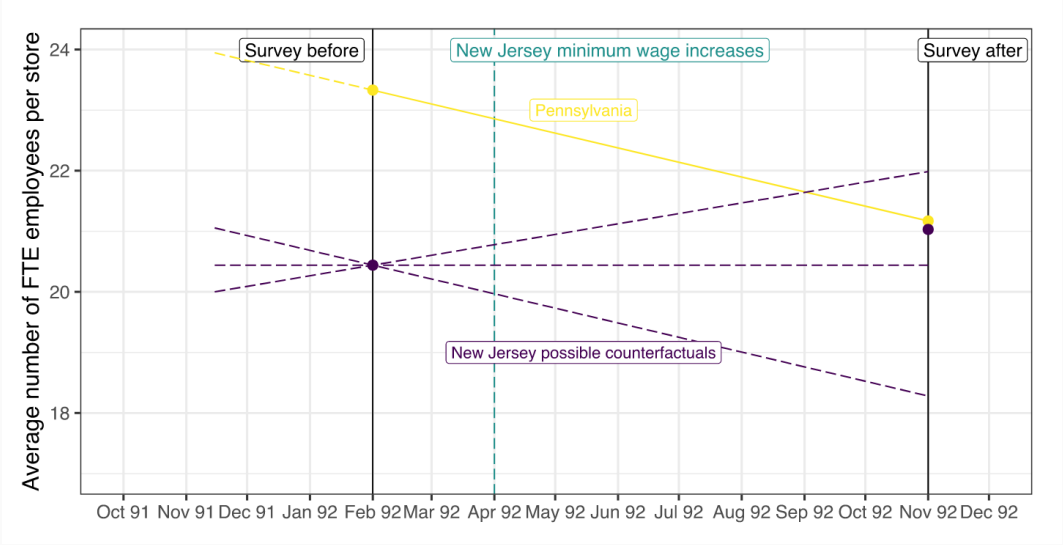
Parallel Trends Assumption

The crucial assumption to identify the causal effect of interest with the DiD estimator is the **Common or Parallel Trends Assumption:**
In the absence of treatment, time trends in the treatment group would have been the same as in the control group.

Example: absent any minimum wage increase, Pennsylvania's fast-food employment trend would have been what we should have expected to see in New Jersey. $\rightarrow$ trend in PA is a reliable counterfactual for NJ

It is impossible to validate whether the assumption holds. (By definition, we cannot observe what would have happened in the absence of treatment in the TG)
Intuitive check: compare trends *before* policy change

Parallel Trends Assumption



Diff-in-Diff Estimator: Regression Form

Diff-in-Dif Estimator: Regression Form

In practice, the DiD estimator is usually estimated in a regression format.

One can show that the DiD estimate obtained by differencing the sample means is the same as the OLS estimate of β_{TE} in

$$Y_{it} = \alpha + \gamma Treat_i + \delta Post_t + \beta_{TE}(Post_t \times Treat_i) + \varepsilon_i$$

$Treat_i$ Treatment group dummy, with index i for cross-sectional units (e.g. individual stores)

$Post_t$ Treatment periods dummy, with index t for time periods

$Treat_i \times Post_t$ Interaction term $\rightarrow$ the coefficient β_{TE} is the DiD causal effect!

Diff-in-Dif Estimator: Regression Form

In practice, the DiD estimator is usually estimated in a regression format.

One can show that the DiD estimate obtained by differencing the sample means is the same as the OLS estimate of β_{TE} in

$$Y_{it} = \alpha + \gamma Treat_i + \delta Post_t + \beta_{TE}(Post_t \times Treat_i) + \varepsilon_i$$

$Treat_i$ Treatment group dummy, with index i for cross-sectional units (e.g. individual stores)

$Post_t$ Treatment periods dummy, with index t for time periods

$Treat_i \times Post_t$ Interaction term $\rightarrow$ the coefficient β_{TE} is the DiD causal effect!

Example:

$$Treat_i = \begin{cases} 0 \text{ if } i \text{ in Pennsylvania,} \\ 1 \text{ if } i \text{ in New Jersey} \end{cases} \qquad Post_t = \begin{cases} 0 \text{ if } t < \text{ April 1992,} \\ 1 \text{ if } t \geq \text{ April 1992} \end{cases}$$

Understanding the Regression

$$Y_{it} = \alpha + \gamma Treat_i + \delta Post_t + \beta_{TE}(Post_t \times Treat_i) + \varepsilon_i$$

$$E(Y_{it}|Treat_i = 0, Post_t = 0) = \alpha$$

$$E(Y_{it}|Treat_i = 0, Post_t = 1) = \alpha + \delta$$

$$E(Y_{it}|Treat_i = 1, Post_t = 0) = \alpha + \gamma$$

$$E(Y_{it}|Treat_i = 1, Post_t = 1) = \alpha + \gamma + \delta + \beta_{TE}$$

$$\underbrace{[E(Y_{it}|Treat_i = 1, Post_t = 1) - E(Y_{it}|Treat_i = 1, Post_t = 0)]}_{\delta+\beta_{TE}=\text{time trend + treatment effect}} -$$

$$\underbrace{[E(Y_{it}|Treat_i = 0, Post_t = 1) - E(Y_{it}|Treat_i = 0, Post_t = 0)]}_{\delta=\text{time trend}} = \beta_{TE}$$

The control group allows to estimate the time trend.

Diff-in-Diff Estimator: Regression Form

The regression form allows to use more than just two time periods (i.e. multiple periods before and/or after the treatment).

Moreover, one can add additional regressors that measure pre-treatment characteristics to the regression:

$$Y_{it} = \alpha + \gamma Treat_i + \delta Post_t + \beta_{TE}(Post_t \times Treat_i) + \pi X_i + \varepsilon_i$$

The additional regressors may lead to a more precise estimate of β_{TE}.

Practice

Practice

Evaluate effect of installation of a garbage incinerator on prices of nearby houses.
Did the installation in 1981 cause house prices to drop?

1. why might a simple comparison of prices of nearby houses before vs. after 1981 not reveal the causal effect of the incinerator?
2. open dataset `"diff_in_diff.csv"`
3. generate a POST dummy that takes value 1 if year = 1981 and 0 if year = 1978
4. generate a TREAT dummy that takes value 1 if house is closer than 5km from the incinerator, using the variable *dist* (distance to incinerator in meters)
5. compute sample mean prices for treatment and control houses before and after the treatment
6. compute the diff-in-diff estimator in its basic form. What does it tell you?
7. formulate the common trends assumption for this analysis
8. obtain the diff-in-diff estimator in a regression framework